# Software Requirements Specification: Engine Control Demand Characterizer

Aaron Willcock

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# 1 Introduction

## 1.1 Purpose

This software aims to:

1. Define a simplified engine model (e.g., accel, speed parameters),
2. define a set of Adaptive Variable-Rate (AVR) tasks as proposed by Biondi et al. [?] (e.g., transition speeds, execution times),
3. combine the engine model and AVR task to form a real-time AVR-based engine control model, hereafter AEM (AVR Engine Model),
4. calculate the maximum demand (see definition CITE) a given AEM can generate, and
5. support result synthesis and visualization (e.g. defining experiment classes and parameters, exporting test data into portable files, enabling comparing algorithm runtimes, etc.)

The intended use is demand characterization for generated AEMs. In furtherance of this intended use, the software also aims to:

1. enumerate multiple calculation options (e.g., different methods),
2. allow algorithm parallelization (where possible),
3. allow approximation (where possible).

## 1.2 Scope

The scope of this software includes three primary areas:

1. model definition and generation,
2. algorithm implementation,
3. parallelization support, and
4. synthesis and data visualization.

For model definition and generation, scope includes:

1. enumeration of existing models and
2. randomized generation based on existing models.

Excluded items include:

1. creation of new engine models and
2. parameter extension beyond existing models.

For algorithm implementation, scope includes:

1. implementation of existing algorithms and
2. parameterization of algorithm variations where feasible (e.g., parameterization of constants in existing research).

Excluded items include:

1. design and implementation of new algorithms and
2. approximation of algorithms no already specified.

For parallelization support, scope includes:

1. multi-threaded implementations of existing algorithms and
2. multi-core implementations of existing algorithms.

Excluded items includes:

1. design or implementation of new parallelization techniques and
2. scripting support for non-SLURM HPC grids.

For synthesis and data visualization, scope includes:

1. definition and implementation of experiment templates,
2. parameterization of experiment variables,
3. design and implementation of experiment test harness(es) for exporting key performance indicators (e.g., demand calculated, runtime, RAM use, etc.) , and
4. definition and use of common key performance data export format (e.g., CSV enumeration of key performance indicators).

## 1.3 Definitions, Acronyms, Abbreviations

- AVR - Adaptive Variable Rate - The Adaptive Variable Rate task model by Biondi et al. CITE
- HPC - High Performance Computing - High performance computing, usually a grid, cluster, or collection of compute devices
- ICE - Internal Combustion Engine - Heat engine using fuel combustion to cause high-temperature, high-pressure gas expansion for performing work
- SI ICE - Spark Ignition ICE - Spark ignition ICE

### 1.4 References

### 1.5 Overview

## 2 General Description

### 2.1 Product Perspective

### 2.2 Product Functions

### 2.3 User Classes and Characteristics

### 2.4 Operating Environment

### 2.5 Design, Implementation, and Environmental Constraints

### 2.6 Assumptions and Dependencies

## 3 Specific Requirements

### 3.1 Functional Requirement 1

### 3.2 Functional Requirement ...

### 3.3 Functional Requirement Z

### 3.4 External Interface Requirements

#### 3.4.1 User Interfaces

#### 3.4.2 Hardware Interfaces

#### 3.4.3 Software Interfaces

### 3.5 Non-Functional Requirements / Performance Requirements

## 4 Acknowledgements

This work templated from IEEE 830-1984 - IEEE Guide for Software Requirements Specifications [1].

# 5 Appendices

## References

[1] IEEE. IEEE Guide for Software Requirements Specifications. *IEEE Std 830-1984*, pages 1–26, Feb. 1984.